

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course : Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTMC402 - THEORY OF MACHINES-I

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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- 1 A kinematic chain is formed when:
 - a) Two links are joined
 - b) Four or more links are joined such that motion of one causes predictable motion of others
 - c) Only three links are used
 - d) Motion is constrained fully
- Gruebler's criterion is used to determine:
 - a) Friction in mechanisms
 - b) Degree of freedom of mechanism
 - c) Length of links
 - d) Material of links
- 3 The number of instantaneous centers in a four-bar mechanism is:
 - a) 2
 - b) 4
 - c) 6
 - d) 5
- 4 The friction in which there is no lubrication is called:
 - a) Boundary friction
 - b) Dry friction
 - c) Fluid friction
 - d) Rolling friction
- 5 Viscosity is defined as:
 - a) Resistance to motion
 - b) Internal resistance of fluid to flow
 - c) Pressure exerted by fluid
 - d) Capillary action

- 6 Torque transmitted by a clutch increases with:
- Decreasing radius
 - Decreasing friction
 - Increasing axial load
 - Decreasing area
- 7 Dynamometers are used to measure:
- Pressure
 - Speed
 - Torque and power
 - Force
- The term "jump" in cam refers to:
- Sudden fall
 - Loss of contact
 - Maximum lift
 - Load transfer
- 9 Pressure angle in cam is:
- Between follower and vertical
 - Between cam profile and follower
 - Between tangents at point of contact
 - Between direction of follower motion and normal to cam surface
- 10 Direct and reverse crank method is used in:
- Gear analysis
 - Engine balancing
 - Cams
 - Brakes
- In locomotive engines, partial balancing is used to:
- Completely eliminate unbalanced forces
 - Avoid all friction
 - Reduce hammer blow
 - Increase crank radius
- 12 Which method is used for velocity analysis in planar mechanisms?
- Vector method
 - Analytical method
 - Instant center method
 - All of the above

Q2. Solve the following.

A) Locate all the instantaneous centres for a four bar mechanism as shown in Fig-1. 6
The lengths of various links are : $AD = 125$ mm ; $AB = 62.5$ mm ; $BC = CD = 75$ mm and the crank AB makes an angle of 60° . If the link AB rotates at a uniform speed of 10 r.p.m. in the clockwise direction, find the angular velocity of the links BC and CD .

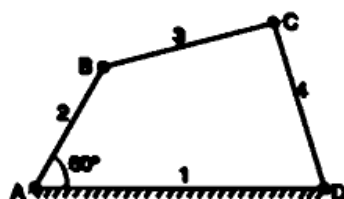


Fig-1

B) Explain with neat sketch 'single slider crank chain'

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Q3. Solve the following.

- A) Derive relation for torque required to overcome friction in case of Flat Pivot Bearing considering uniform pressure theory and uniform wear theory
- B) In a thrust bearing, the external and the internal diameters of the contacting surfaces are 320 mm and 200 mm respectively. The total axial load is 80 kN and the intensity of pressure is 350 kN/m^2 . The shaft rotates at 400 rpm. Taking the coefficient of friction as 0.06, calculate the power lost in overcoming the friction. Also, find the number of collars required for the bearing.

Q4. Solve Any Two of the following.

- A) The inner and outer radii of a single plate clutch are 40 mm and 80 mm respectively. Determine the minimum, maximum and average pressure when the axial force is 3 kN.
- B) A band brake acts on the $\frac{3}{4}$ th of circumference of a drum of 450 mm diameter which is keyed to the shaft. The band brake provides a braking torque of 225 N-m. One end of the band is attached to a fulcrum pin of the lever and the other end to a pin 100 mm from the fulcrum. If the operating force is applied at 500 mm from the fulcrum and the coefficient of friction is 0.25, find the operating force when the drum rotates in the anticlockwise direction.
- C) Explain with neat sketch the Rope Brake Dynamometer. Also derive relation for the engine brake power.

Q5. Solve Any Two of the following.

- A) A cam, with a minimum radius of 25 mm, rotating clockwise at a uniform speed is to be designed to give a knife edge follower, at the end of a valve rod, motion described below:
 - i. To raise the valve through 50 mm during 120° rotation of the cam
 - ii. To keep the valve fully raised through next 30°
 - iii. To lower the valve during next 60°
 - iv. To keep the valve closed during rest of the revolution i.e. 150° ;The diameter of the cam shaft is 25 mm. Draw the profile of the cam when the line of the stroke is offset by 16 mm from the axis of the cam shaft. The displacement of the valve, while being raised and lowered, is to take place with simple harmonic motion. Determine the maximum acceleration of the valve rod when the cam shaft rotates at 100 r.p.m.
- B) Explain Displacement, Velocity and Acceleration diagrams when the follower moves with Cycloidal Motion
- C) With neat sketch explain following types of cam with neat sketch
 - i) Globoidal cam (convex cam and concave)
 - ii) R-R-R cam

Q6. Solve Any Two of the following.

- A) Explain primary and secondary unbalanced forces in case of reciprocating engine

- B) A shaft carries four masses P, Q, R and S of magnitude 200 kg, 300 kg, 400 kg and 600 kg respectively and revolving at radii 80 mm, 70 mm, 60 mm, and 80 mm in planes measured from P at 300 mm, 400 mm and 700 mm. The angles between the cranks measured anticlockwise are P to Q 45° , Q to R 70° and R to S 120° . The balancing masses are to be placed in planes X and Y. The distance between the planes P and X is 100 mm, between X and Y is 400 mm and between Y and S is 200 mm. If the balancing masses revolve at a radius of 100 mm, find their magnitudes and angular positions.
- C) Explain direct and reverse crank method of radial engine balancing

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